ETH
Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

Institute for Dynamic Systems and Control
IDSC
Institut für Dynamische Systeme und Regelungstechnik

Exam **February 6th, 2020**

Signals & Systems **(151-0575-01)** **Dr. A. Carron**

Exam

Student number:	First name and last name:
.....	.....

Exam Duration:	**150 Min**
Number of Problems:	**11**
Number of Points:	**67**
Permitted aids:	**Notes: A single A4 sheet of paper (double-sided; hand-written or computer-typed)**
Important:	Multiple versions of the exam exist. The order of the answers has been permuted randomly. Questions are not equally weighted. Answers must be justified, except where otherwise stated. Use only the paper provided for your calculations; additional paper is available from the supervisors. Note that multiple choice questions have only one correct answer. No negative points will be given for incorrect answers.

Good luck!

Corrected

Question 1 **8 Points**

Consider the following causal continuous-time system

$$\dot{y}(t) = ay(t) + u(t),$$

for which the corresponding transfer function results in

$$H(s) = \frac{1}{s-a}.$$

Assume that $u(t)$ is piece-wise constant over intervals T_s:

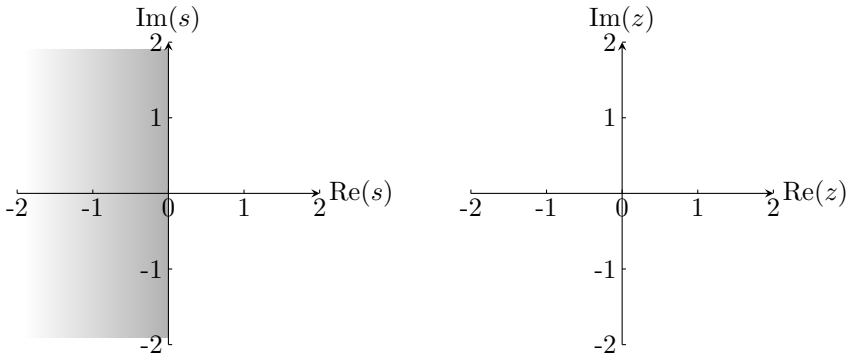
$$u(t) = u[n],\ nT_s \leq t < (n+1)T_s.$$

The system is discretized such that

$$y[n] = (1 + aT_s)y[n-1] + T_s u[n-1],$$

where $y[n] = y(nT_s)$ and $u[n] = u(nT_s)$.

1. Which discretization method is used to obtain the discrete-time system?
2. Based on the continuous-time system and its discretized version, determine the z-Transform variable as a function of s, i.e. a relation of the form $z = f(s)$.
3. A plot of the region of stability in the continuous domain is given below. Using the previously found relation, plot the region in the discrete domain to which stable poles in the continuous domain are mapped. Provide an explanation of how you obtain this region.



Solution:

1. The discretization method used is forward Euler. **2 Points**
2. The z-Transform of the discretized system is

   $$Y(z) = (1 + aT_s)z^{-1}Y(z) + T_s z^{-1}U(z),$$

   which provides the following transfer function

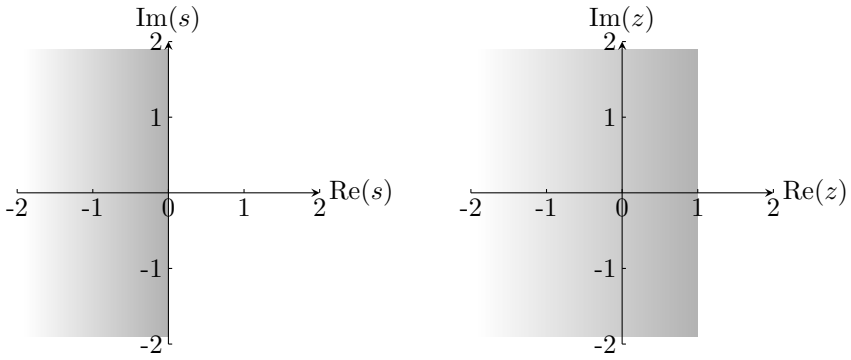
   $$H(z) = \frac{1}{\frac{z}{T_s} - \frac{(1+aT_s)}{T_s}}.$$

   By matching the denominators of $H(s)$ and $H(z)$, we obtain

   $$\frac{z}{T_s} - \frac{(1+aT_s)}{T_s} = s - a \ \Rightarrow\ z = 1 + sT_s.$$

   One could also simply notice that $z = e^{sT_s} \approx 1 + sT_s$ using forward Euler approximation. **3 Points**

3. The region of stability in the continuous domain corresponds to $\mathrm{Re}(s) < 0$. Given the relation found in 2., we can show that in the discrete domain, stable poles in the continuous domain are mapped to $\mathrm{Re}(\frac{z-1}{T_s}) < 0 \;\Rightarrow\; \mathrm{Re}(z) < 1$



3 Points
Grading Remarks:

- **1 Point** for correct plot.
- **2 Points** for correct argument

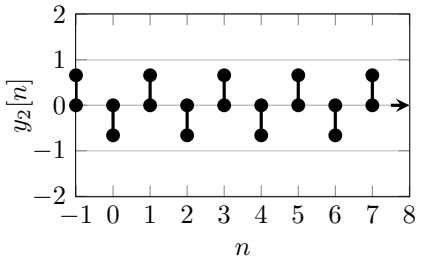
Question 2 **2 Points**

Consider a signal $u[n] = 1 + \cos(\pi n)\ \forall n$ processed by the following system

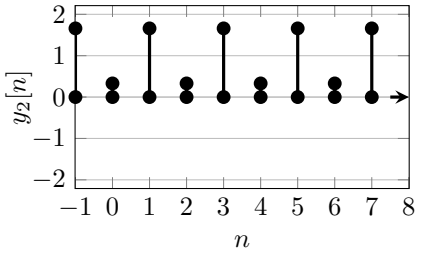
$$y_1[n] = y_1[n-1] + 0.5(u[n] - y_1[n-1])$$
$$y_2[n] = u[n] - y_1[n].$$

Which of the following plots is the output $y_2[n]$? No explanation is needed.

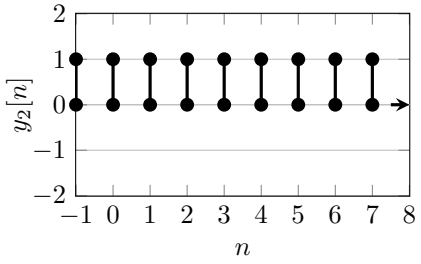
■



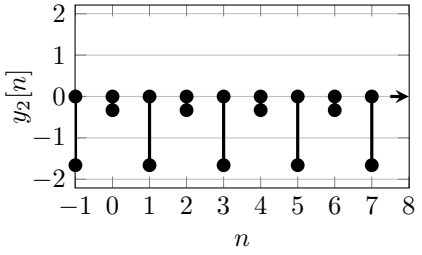
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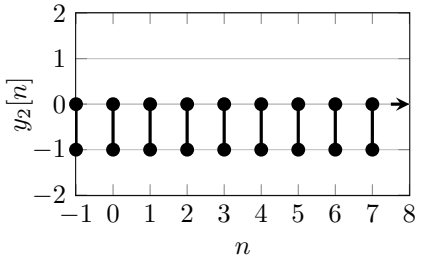
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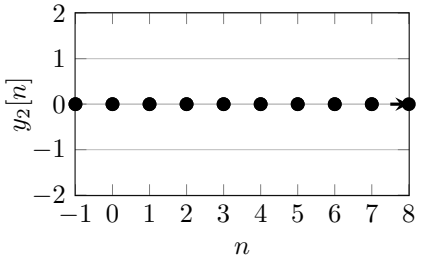
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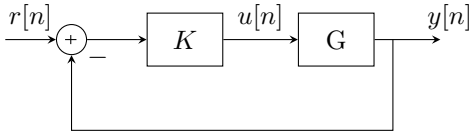
Solution:
The filter is a simple high-pass first order filter and eliminates the constant. So the correct answer is the one oscillating around zero. (**3 Points**)

Question 3 **8 Points**

Consider a causal, unstable LTI system G with transfer function

$$H(z) = \frac{b}{1 + az^{-1}},$$

which is stabilized with a feedback-loop using a constant gain $K = 3$ as depicted in the following illustration.



1. To identify the coefficients of the transfer function $H(z)$, the signals $r_1[n] = 1$ and $r_2[n] = (-1)^n\ \forall n$ are given as reference to the closed-loop system. The measured outputs are $y_1[n] = \frac{1}{2}$ and $y_2[n] = \frac{3}{2}(-1)^n$, respectively. Calculate the values of a, b.
2. Given $b = 2$, for which values of a is the closed-loop system BIBO stable?

Solution:
The closed loop transfer function is

$$W(z) = \frac{KH(z)}{1 + KH(z)} = \frac{Kb}{1 + \frac{Kb}{1+az^{-1}}} = \frac{Kb}{az^{-1} + (1 + Kb)}.$$ (**2 Points**)

Part 1: We can write a system of equations and find the 2 parameters of the system. (**2 Points**)

$$\begin{cases} W(1) = \frac{1}{2} \\ W(-1) = \frac{3}{2} \end{cases} \implies \begin{cases} \frac{3b}{a+(1+3b)} = \frac{1}{2} \\ \frac{3b}{-a+(1+3b)} = \frac{3}{2} \end{cases} \implies \begin{cases} 3b = \frac{1}{2}(a + (1+3b)) \\ b = \frac{1}{2}(-a + (1+3b)) \end{cases} \implies \begin{cases} 6b = a + 1 + 3b \\ 2b = -a + 1 + 3b \end{cases}$$

If we sum up the two equations we get

$$8b = 2 + 6b \implies 2b = 2 \implies b = 1,$$

and consequently from the first equation

$$6 = a + 4 \implies a = 2. \text{ (\textbf{2 Points})}$$

Part 2: If we analyze the denominator of the closed-loop transfer function, we have:

$$az^{-1} + (1+6) = az^{-1} + 7 \implies z = \frac{a}{7}$$

So in order to guarantee BIBO stability $|\frac{a}{7}| < 1 \Rightarrow |a| < 7$. (**2 Points**)

Question 4 **10 Points**

Consider the signal

$$u[n] = a_0 + a_1 \cos(\frac{\pi}{2}n) + a_2(-1)^n, \quad n = \{\dots, -1, 0, 1, \dots\},$$

where $a_0, a_1, a_2 \in \mathbb{R}$ are fixed, but unknown constants.
Design a causal, stable, LTI filter G so that if u has the form above and $y = Gu$, then

$$y[n] = a_0 + 2a_2(-1)^n \qquad n = \{\dots, -1, 0, 1, \dots\}.$$

Solution:

$$u[n] = a_0 e^{j\Omega_0 n} + a_1 \left(\frac{e^{j\Omega_1 n} + e^{-j\Omega_1 n}}{2} \right) + a_2 e^{jj\Omega_2 n},$$

where $\Omega_0 = 0$, $\Omega_1 = \frac{\pi}{2}$, and $\Omega_2 = \pi$. **1 Point**.
So all we need to do is find a transfer function $H(z)$ for G such that

$$H(z = e^{j\Omega_0}) = 1, \quad H(z = e^{j\Omega_2}) = 2,$$
$$H(z = e^{j\Omega_1}) = 0, \quad H(z = e^{-j\Omega_1}) = 0$$

1 Point for each condition = 4 Points.

If we can make this a FIR transfer function, it is guaranteed to be stable **1 Point**. Causality can easily be imposed by multiplying by z^{-N}, for a sufficiently large N **1 Point**.

$$\begin{aligned} H_1(z) &= \alpha(z - e^{j\Omega_2})(z - e^{-j\Omega_2}) \\ &= \alpha(z^2 - z(e^{i\Omega_2} + e^{-i\Omega_2}) + 1) \\ &= \alpha(z^2 - z(2\cos(\Omega_2)) + 1) \\ &= \alpha(z^2 + 1). \end{aligned} \tag{1}$$

$$H_1(z = e^{i\Omega_0}) = H_1(z = e^{i\Omega_2}) = 1 \implies \alpha = \frac{1}{2}$$

We now design a first order filter to adjust the gains at Ω_0 and Ω_2. **(2 Points), one per condition**.

$$H_2(z) = \alpha + \beta z \implies \begin{cases} \alpha + \beta = 1 \\ \alpha - \beta = 2 \end{cases} \implies \begin{cases} \alpha = -\frac{1}{2} \\ \beta = \frac{3}{2}. \end{cases}$$

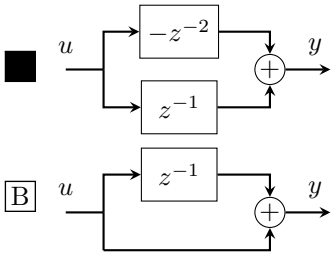
The wanted transfer function becomes

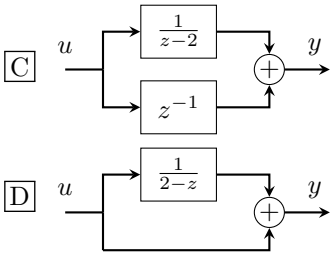
$$H(z) = H_1(z)H_2(z) = \alpha(z^2+1)(\alpha + \beta z)z^{-3} = \frac{1}{2}(z^2+1)(\frac{3}{2}z - \frac{1}{2})z^{-3}$$

where z^{-3} has been added to make the filter causal. **1 Point**.

Question 5 **2 Points**

Let $\{u[n]\} = \{ns[n-1]\}$ be the input sequence applied to the causal LTI system G. Which of the following plots could represent the transfer function of G if the output is equal to $\{y[n]\} = \{s[n-2]\}$? No explanation is needed.





Solution:
We can notice that the output is the derivative of the input shifted of one time step. So the impulse response is $\{h[n]\} = \{\delta[n-1] - \delta[n-2]\}$. If we use transfer functions, we have that the transform of $S(z) = Z(s[n-1]) = z^{-1}\frac{z}{z-1} = \frac{1}{z-1}$, applying the differentiation proprieties, we get

$$U(z) = -z\frac{dS(z)}{dz} = -z\frac{-1}{(z-1)^2} = \frac{z^{-1}}{(1-z^{-1})^2}.$$

The Z-transform of the output is

$$Y(z) = z^{-2}\frac{1}{1-z^{-1}}.$$

So the transfer function is

$$H(z) = \frac{Y(z)}{U(z)} = \frac{z^{-2}}{1-z^{-1}}\frac{(1-z^{-1})^2}{z^{-1}} = z^{-1} - z^{-2}.$$

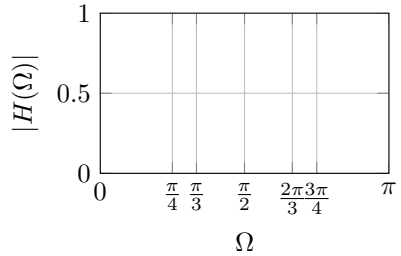
2 Points.

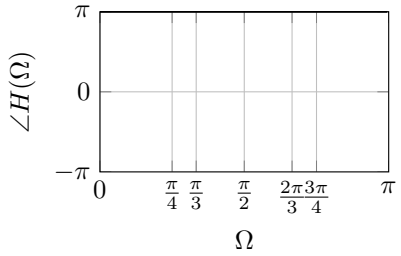
Question 6 **6 Points**

Consider the following transfer function

$$H(z) = 0.25(z + 2 + z^{-1}).$$

1. Plot magnitude and phase response of the transfer function $H(z)$.





2. Determine if the system is FIR or IIR, causal or non-causal.
3. Given the input signal $u[n] = \cos(\frac{\pi}{3}n)$, determine the output $y[n]$.

Solution:

1. The frequency response is $H(\Omega) = \frac{1}{2}(1 + \cos(\Omega))$ **2 Points**
2. The filter is a FIR moving average non-causal filter. **2 Points, one per each condition**
3. We see that $H(z) = \frac{3}{4}$, so the output is $y[n] = \frac{3}{4}\cos(\frac{\pi}{3}n)$. **2 Points**

Question 7 **7 Points**

Consider a causal LTI system G, for which we observe that given an input

$$\{u[n]\} = \{s[n]\} - \{s[n-2]\},$$

the corresponding output is

$$\{y[n]\} = G\{u[n]\} = \{\delta[n-1]\} + \{\delta[n-4]\}.$$

1. Determine the impulse response sequence $\{h[n]\}$.
2. Determine whether system G is BIBO stable.

Solution:

1. First notice that the input can be rewritten as

$$\{s[n]\} - \{s[n-2]\} = \{s[n]\} - \{s[n-1]\} + \{s[n-1]\} - \{s[n-2]\} = \{\delta[n]\} + \{\delta[n-1]\},$$

or, alternatively, notice after plotting the input that it is equal to the sum of unit impulses. This means that

$$\{\delta[n-1]\} + \{\delta[n-4]\} = G(\{s[n]\} - \{s[n-2]\}) = G(\{\delta[n]\} + \{\delta[n-1]\}) = \{h[n]\} + \{h[n-1]\}.$$

From the difference equation $h[n] = -h[n-1] + \delta[n-1] + \delta[n-4]$, and given that the system is causal, we obtain

$$h[n] = \{0, 1, -1, 1, 0, 0, \dots\} \quad n \geq 0.$$

6 points
Grading Remarks:

- **4 Points** for convincing procedure for obtaining $h[n]$.
- **2 Points** for correct signal computation.

2. Check the stability condition on the impulse response computed in 1.

$$\sum_{n=-\infty}^{n=+\infty} |h[n]| = 3 < \infty \;\Rightarrow\; \text{the system is stable}$$

1 Point

Question 8 **4 Points**

Consider an LTI system G, with $G \neq 0$. Mark the statements that are guaranteed to be true without providing an explanation.

1. If u is a complex exponential sequence with frequency Ω, then $y = Gu$ has the same frequency as u.

   ■ True B False

2. If $u[n] = \cos(\Omega_1 n) + \cos(\Omega_2 n)$ is the input to system G, and $\Omega = \Omega_1 + \Omega_2$, the output of system G is:

   $y[n] = |H(\Omega)| \cos(\Omega n + \Theta_H(\Omega))$.

   C True ■ False

3. If $u[n] = e^{j\Omega n}$ is the input to system G, and G is determined by the cascade of two LTI systems G_1 and G_2, the output of system G is:

   $y[n] = |H_1(\Omega)||H_2(\Omega)|e^{j(\Omega n + \Theta_{H_1}(\Omega) + \Theta_{H_2}(\Omega))}$.

   ■ True F False

4. If $u[n] = -1^n + \cos(\Omega n)$ is the input to system G, the output is:

   $y[n] = |H(\Omega)| \cos(\Omega n + \Theta_H(\Omega)) + |H(0)| \cos(\Theta_H(0))$.

   G True ■ False

Solution:
T (**1 Point**), F (**1 Point**), T (**1 Point**), F (**1 Point**)

Question 9 **8 Points**

Consider the following continuous-time periodic signals

$$x_i(t) = \cos(w_i t + \frac{\pi}{4}), \quad w_i \geq 0,$$

with $i = 1, 2, 3$. Starting at $t = 0$, we record one period for each signal of length T_i, and then uniformly sample it using sampling time T_{s_i}. For each signal we have:

a. $T_1 = 3$, $T_{s_1} = 1$,

b. $T_2 = 16$, $T_{s_2} = 5$,

c. $T_3 = 16$, $T_{s_3} = 9$.

1. Determine the DFS coefficients $X_i[k]$ of the sampled signals for $i = 1, 2, 3$.
2. Determine if aliasing is occurring for $i = 1, 2, 3$.

Solution:

1. The discrete-time signal can be obtained by noticing that

   $$w_i = |w_i| = \frac{2\pi}{T_i},$$

   and that

   $$w_i t = \frac{2\pi}{T_i} t = 2\pi \frac{T_{s_i}}{T_i} n = \Omega_i n.$$

   This means that the sampled signal will be $x_i[n] = \cos(2\pi \frac{T_{s_i}}{T_i} n + \frac{\pi}{4})$. The period of each signal can be determined by using

   $$\frac{\Omega_i}{2\pi} = \frac{T_{s_i}}{T_i} = \frac{m_i}{N_i},$$

   providing the fundamental period N_i. We can now determine the DFS coefficients for each case

a. $\frac{T_{s_1}}{T_1} = \frac{1}{3} \Rightarrow N_1 = 3, x_1[n] = \cos(\frac{2\pi}{3}n + \frac{\pi}{4}) = \frac{1}{2}(e^{j\frac{\pi}{4}}e^{j\frac{2\pi}{3}n} + e^{-j\frac{\pi}{4}}e^{-j\frac{2\pi}{3}n}) = \frac{1}{3}\sum_{n=0}^{2} X_1[k]e^{jk\frac{2\pi}{3}n}$.

We observe that $X_1[0] = 0$, $X_1[1] = \frac{3}{2}e^{j\frac{\pi}{4}}$, $X_1[2] = \frac{3}{2}e^{-j\frac{\pi}{4}}$.

b. $\frac{T_{s_2}}{T_2} = \frac{5}{16} \Rightarrow N_2 = 16, x_2[n] = \cos(\frac{5\pi}{8}n + \frac{\pi}{4}) = \frac{1}{2}(e^{j\frac{\pi}{4}}e^{j\frac{5\pi}{8}n} + e^{-j\frac{\pi}{4}}e^{-j\frac{5\pi}{8}n}) = \frac{1}{16}\sum_{n=0}^{15} X_2[k]e^{jk\frac{2\pi}{16}n}$.

The only non-zero coefficients are $X_2[5] = 8e^{j\frac{\pi}{4}}$, $X_2[11] = 8e^{-j\frac{\pi}{4}}$.

c. $\frac{T_{s_3}}{T_3} = \frac{9}{16} \Rightarrow N_3 = 16, x_3[n] = \cos(\frac{9\pi}{8}n + \frac{\pi}{4}) = \frac{1}{2}(e^{j\frac{\pi}{4}}e^{j\frac{9\pi}{8}n} + e^{-j\frac{\pi}{4}}e^{-j\frac{9\pi}{8}n}) = \frac{1}{16}\sum_{n=0}^{15} X_3[k]e^{jk\frac{2\pi}{16}n}$.

The only non-zero coefficients are $X_3[7] = 8e^{-j\frac{\pi}{4}}$, $X_3[9] = 8e^{j\frac{\pi}{4}}$.

7 Points
Grading Remarks:

- **2 Points** for each DFS.
- **1 Point** for correct discretization.

2. The condition that has to be satisfied is $-\frac{\pi}{T_{s_i}} < w_i < \frac{\pi}{T_{s_i}}$ or $-\pi < \Omega_i < \pi$, which is respected by x_1 ($\Omega_1 = \frac{2\pi}{3}$) and x_2 ($\Omega_2 = \frac{5\pi}{8}$) but not by x_3 ($\Omega_3 = \frac{9\pi}{8}$). **1 Point**

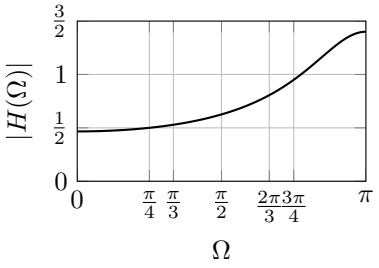
Question 10 **8 Points**

Consider the following discrete-time system G described by

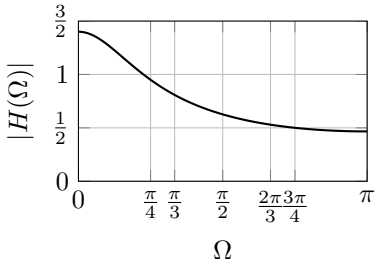
$$y[n] = 0.5y[n-2] + u[n] - u[n-2].$$

1. If $u[n] = 1 + 0.5\sin(\frac{\pi}{2}n) + 0.25\cos(\pi n)$, determine the DFS coefficients of $y[n]$.
2. Determine the magnitude response $|H(\Omega)|$ of system G without providing an explanation.

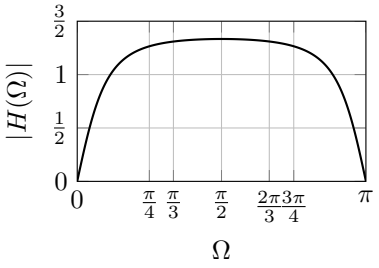
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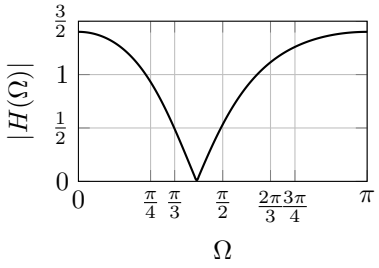
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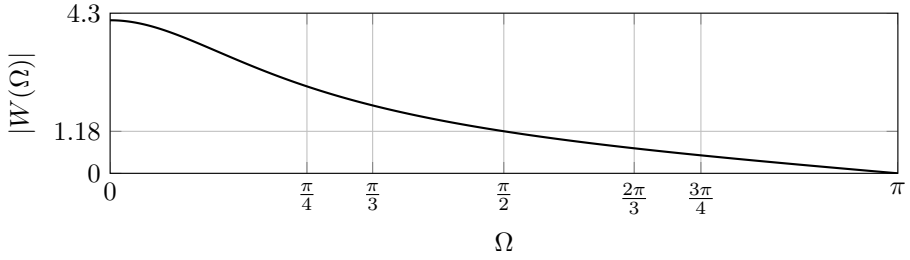
■



D



3. Consider the low-pass filter G_{LP}, for which the magnitude response is



Assume that its z-Transform is defined as $W(z)$. Determine which of the following approximations is true. No explanation is needed.

E $H(z) \approx W(z)$
F $H(z) \approx 1 - W(-z)$
G $H(z) \approx W(z)(1 - W(z))$
■ $H(z) \approx W(z)W(-z)$
I $H(z) \approx 1 - W(z)$
J $H(z) \approx W(-z)$

Solution:

1. Frequency response is $H(\Omega) = \frac{1-e^{-j2\Omega}}{1-0.5e^{-j2\Omega}}$.
   Furthermore $u[n] = e^{j0n} + \frac{0.25}{j}(e^{j\frac{\pi}{2}n} - e^{-j\frac{\pi}{2}n}) + 0.125(e^{j\pi n} + e^{-j\pi n}) = \frac{1}{4}\sum_{n=0}^{3} U[k]e^{jk\frac{2\pi}{4}n}$.
   Then $U[0] = 4$, $U[1] = -j$, $U[2] = 0.5$, $U[3] = j$. We know that $Y[k] = H(e^{jk\frac{2\pi}{4}})U[k]$ which implies that $Y[0] = 0$, $Y[1] = -\frac{4}{3}j$, $Y[2] = 0$, $Y[3] = \frac{4}{3}j$. (**3 Points**)
2. Correct answer is B. (**2 Points**)
3. Correct answer is H. (**3 Points**)

Question 11 **4 Points**

Consider a vector $y \in \mathbb{C}^N$ collecting the discrete-time sequence $\{x[n]\},\ n = 0, \ldots, N-1$ and a vector $b \in \mathbb{C}^N$ collecting the corresponding DFT coefficients $\{X[k]\},\ k = 0, \ldots, N-1$. The following relation

$$y = \Phi b,$$

where $\Phi \in \mathbb{C}^{N \times N}$, enables reconstruction of the whole discrete-time sequence when considering all DFT coefficients simultaneously, or vice-versa.

1. Determine the entries $[\Phi]_{il},\ i = 0, \ldots, N-1,\ l = 0, \ldots, N-1$, so that the DFT transform holds.
2. Show that linearity holds.

Solution:

1. $[\Phi]_{ij} = \frac{1}{N} e^{ji\frac{2\pi}{N}l}$ **2 Points**
2. $y_1 = \Phi b_1$, $y_2 = \Phi b_2$ and $b = \alpha_1 b_1 + \alpha_2 b_2$. Then $y = \Phi b = \Phi(\alpha_1 b_1 + \alpha_2 b_2) = \alpha_1 \Phi b_1 + \alpha_2 \Phi b_2 = \alpha_1 y_1 + \alpha_2 y_2$. **2 Points**